From Detection to Attribution: Linking Two Decades of Antarctic Stream-Channel Change to Its Climate Drivers, from Terrain Data Alone

Summary. This project turns an existing Antarctic stream-change monitor into a tool that explains change. Barlow (2026) mapped two decades of stream-channel change across the McMurdo Dry Valleys; the proposed work asks the next questions. (O1) Which climate drivers control that change? (O2) What is changing geomorphologically outside the channels — across the whole valley floor — and do different kinds of change answer to different drivers? (O3) Can every change map carry an honest, per-pixel error bar?

1. Background and the Gap

The McMurdo Dry Valleys (MDVs) are Earth's largest ice-free Antarctic desert and, for a century, were considered the planet's most geomorphologically stable landscape. That assumption is breaking down. The system is energy-limited — there is plenty of frozen water but barely enough heat to melt it — so even small increases in absorbed sunlight, together with thawing permafrost, translate directly into ephemeral summer streams: channels that flow for only a few weeks each year, move sediment while they do, and reshape themselves in the process. These streams are the most sensitive measurable indicator of climate change in the most stable place on Earth, a continental-scale "climate canary."

Barlow (2026) [1] established the measurement foundation in three transferable pillars:

1. Terrain-only semantic segmentation: a U-Net delineates stream boundaries from DEM derivatives alone (elevation, slope, aspect most informative; no water/color signal), so it works in channels that are dry 11 months a year. 2. Satellite-DEM validation: free REMA satellite DEMs are proven accurate enough for centimeter-to-meter stream-corridor change detection, using ICP (iterative closest point — sliding one 3-D surface onto another until they match) for alignment and NMAD (normalized median absolute deviation — a standard deviation that ignores outliers) for honest error bars; the errors follow a Laplacian (sharp-peaked, heavy-tailed) distribution rather than a Gaussian, which is why the robust statistic is required. 3. Multi-epoch change detection: subtract one epoch's DEM from another (DEM-of-Difference, DoD) inside the segmented stream masks, and count only elevation changes exceeding the level of detection (LOD95 = 1.96 × NMAD — the smallest change distinguishable from measurement noise at 95% confidence). This yields per-stream erosion/deposition and acceleration rates across four valleys and three epochs (2001 & 2014 lidar, 2021–23 REMA).

The gap. Barlow's dissertation ends at *description*, and at the channel edge. It maps where in-channel change is happening (Denton Hills is the erosion hotspot; parts of Taylor Valley are accelerating), but it explicitly defers coupling that change to physical/climate drivers to future work — and its change analysis is masked to the stream channels, so everything the elevation differencing measures *outside* them (thawing ground, slopes, fans, lake margins) is discarded unexamined. A Future Investigator project lives precisely in that gap: explain the change, and stop throwing most of the measurement away. This is a genuine scientific increment, not a replication.

Relevance to NASA. The 2001 baseline epoch is NASA data — an ATM airborne lidar campaign — that has never been fully exploited for surface change. The proposed work converts that NASA archive, the free REMA satellite record, and reanalysis into a continuing, uncertainty-calibrated measurement of climate-driven surface change in the Dry Valleys, serving the Earth Science Division's cryosphere and Earth-surface-change focus areas. The methodological deliverables generalize: terrain-only feature detection and calibrated per-pixel change thresholds (O3) apply to any elevation-differencing record, including those produced by ICESat-2 and NISAR. The MDVs are also the canonical terrestrial analog for cold-desert planetary surfaces — a long-standing NASA investment this project extends with open, reusable tools. The FI development itself is Division-aligned: open geospatial data, machine learning on remote sensing, and honest uncertainty quantification.

2. Objectives and Hypotheses

#	Objective	Hypothesis	Increment over Barlow
O1	Attribution (headline science). Match each stream's measured change rate against candidate drivers — temperature (as positive degree-days, PDD: a running total of melting weather), sunlight, discharge, glacier mass balance, thaw depth — from the MDV-LTER network plus ERA5/AMPS.	H1: *cumulative melt energy*, not water volume alone, controls sediment movement — a heat-based model predicts each stream's rate better than discharge and explains the scatter discharge leaves behind.	Listed verbatim as Barlow future work; never quantified.
O2	Landscape-wide geomorphic change. Extend change detection from the channel masks to the full valley-floor surface; classify significant change by process type — channel migration, thaw-driven subsidence (thermokarst), slope movement, fan/delta growth, lake-margin shift; attribute each type to its drivers.	H2: different process types answer to different drivers — channel change tracks melt energy and discharge, while thermokarst tracks summer thaw depth. Each type should carry a distinct driver fingerprint.	Barlow's change analysis is confined to the detected channel outlines; the rest of the DoD measurement is discarded.
O3	Calibrated uncertainty. Replace the single valley-wide noise estimate with a per-pixel threshold that accounts for slope, aspect, and sensor.	H3: a 95% threshold built this way behaves like one — on unchanged ground, only ~5% of pixels exceed it.	Barlow uses a single global NMAD per epoch-pair.

Predictive payoff (ties O1–O3 together): a model that, given a driver field, predicts *where acceleration migrates next*, the actionable form of the "climate canary."

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3. Methodology

Data (all free/public; see §5). Three elevation snapshots form the backbone: 2001 airborne lidar (NASA ATM, 2 m), 2014 airborne lidar (NCALM, 1 m), and the REMA satellite DEM (2 m, 2021–23). From each snapshot the pipeline derives the terrain layers the detector reads: elevation, slope, aspect, curvature, flow accumulation (a "where water would go" map), and lidar intensity. Driver data come from the MDV-LTER station network — meteorology, 21 stream-discharge gauges, mass balance for 7 glaciers, and soil temperature/moisture (a proxy for summer thaw depth) — with the ERA5 and AMPS weather models filling the gaps between stations. Training labels: the public MCM-LTER stream centerlines, standing in for Barlow's access-gated 217 hand-digitized tiles (§6 risk).

O1, Attribution. For each gauged stream and epoch: (i) measure the change rate — the DoD inside its channel mask; (ii) build the stream's energy history (PDD, sunlight, discharge, thaw depth) from LTER stations plus ERA5/AMPS; (iii) fit hierarchical regression and random-forest models relating rate to drivers, always validating on streams held out of fitting; (iv) test H1 head-to-head — does melt energy predict rates better than water volume alone? For the Taylor streams with all three epochs, the *change* in rate (acceleration) is compared against the *trend* in each driver.

O2, Landscape-wide change. The DoD already measures the entire surface; the channel mask simply discards most of it. The work will apply the O3 per-pixel thresholds to the full valley floor, extract coherent patches of significant change, and classify them by process type from their morphology and setting (slope position, proximity to channels and lakes, ice-cored terrain). Each type's rates then enter the O1 driver models separately, testing H2's prediction of distinct driver fingerprints. The FI's prior work classifying terrain-disturbance features by type on a completely different landscape (§4) shows this move is in hand.

Sensor continuity (supporting method, not an objective). The record continues past the last lidar flight (2014) only on REMA, and satellite elevations disagree with lidar in terrain-dependent ways (a domain shift). The work will measure that disagreement over unchanged ground as a function of slope and aspect and correct it before any cross-sensor differencing — a prerequisite for O1–O3. Extension beyond Taylor Valley proceeds where REMA quality supports it.

O3, Calibrated uncertainty. The work will replace the single global noise estimate with one that varies by slope, aspect, and sensor, and publish it as a per-pixel detection-threshold map. Calibration will be verified the honest way: on held-out ground that did not change, a 95% threshold should be exceeded by only ~5% of pixels.

Reproducibility/QC. The pipeline is fully scripted; input data and all derivatives regenerate from source, with robust statistics and CRS checks enforced at every step.

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4. FI Qualifications

The FI has independently built and operated this exact toolchain, U-Net semantic segmentation on lidar-derived terrain rasters, ICP co-registration, and DEM-of-Difference change analysis, on an unrelated landscape (channels, roads, and disturbance scars detected from elevation alone in the Appalachian Plateau). That prior work shows the architecture transfers across sensors and biomes, that classifying terrain disturbance by process type (the core move of O2) is demonstrated, and that the FI commands the full pipeline end to end.

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5. Data Requirements and Availability

Every dataset required for the proposed work is free/public, so data availability poses no schedule risk. The single access-gated dependency is Barlow's training labels (mitigated below and in §6).

Dataset (source)	What it shows	Role in the proposed work	Availability
2001 airborne lidar, 2 m (NASA ATM, via USGS)	Valley-floor surface in 2001	Baseline epoch for all change detection	public, free
2014 airborne lidar, 1 m (NCALM, via OpenTopography)	Same terrain 13 yr later at benchmark accuracy	Reference epoch; anchors the error model	public, free
REMA satellite DEM, 2 m, 2021-23 (PGC/Maxar stereo)	Most recent surface, continent-wide	Extends the record past the last lidar flight (sensor-continuity correction, §3)	public, free
Per-epoch terrain derivatives (slope/aspect/curvature/MFD-flow/intensity)	Channel-diagnostic terrain form	Input features the U-Net segmenter reads	computed from the DEMs
LTER stream gauges, 21 streams, daily, 1990s-present (EDI)	Meltwater each stream carried	Direct discharge driver for O1	public, free
LTER meteorology (EDI)	Air temperature, radiation, wind	PDD + insolation melt-energy drivers (O1/H1)	public, free
LTER glacier mass balance, 7 glaciers (EDI)	Seasonal ice gain/loss per glacier	Links stream water supply to its source	public, free
LTER soil temperature/moisture (EDI)	Summer thaw depth	Active-layer driver (O1/H1)	public, free
ERA5 monthly (Copernicus CDS); AMPS 2.67 km (NCAR GDEX)	Continuous modeled atmosphere	Gap-fills drivers between stations and to ungauged streams; AMPS cross-checks ERA5	public, free

Dataset (source)	What it shows	Role in the proposed work	Availability
MCM-LTER stream channel polygons (EDI `knb-lter-mcm.6007`)	Mapped channel of every named stream	Per-stream rate masks + label stand-in	public, free
Barlow's 217 hand-digitized tiles (author-gated)	Expert-traced channel boundaries	Gold-standard U-Net training labels	▲ lone access-gated dependency (see §6)
Cross-valley DEMs, all four valleys (OpenTopography/PGC)	Terrain beyond Taylor Valley	Extension beyond Taylor Valley where coverage allows	public, free
WorldView/Maxar imagery; published rates (PGC restricted; literature)	Visual ground truth; independent rates	Optional label QC / cross-check vs Barlow	optional

6. Management, Timeline, Risks, and Data Management Plan

FI role & development. The FI is intellectual lead and primary author; the advisor (PI of record) provides domain mentorship. Development plan: present at AGU (cryosphere) and an ISAES/SCAR venue; co-author the attribution paper (O1) as a stand-alone publication; complete graduate coursework in Bayesian/hierarchical modeling and remote-sensing uncertainty.

Timeline (3 years).

Period	Milestones
Yr 1	Change-detection chain built from public data; lidar–REMA disagreement measured and corrected; full driver harmonization (PDD/insolation/active-layer series per gauged stream); O1 attribution model on Taylor Valley; calibrated-uncertainty prototype (O3). Deliverable: attribution manuscript drafted.
Yr 2	Landscape-wide change detection + process-type classification on Taylor Valley (O2), gated by the O3 thresholds; per-type driver attribution; AGU presentation. Deliverable: O2 paper.
Yr 3	Predictive model (where acceleration migrates next); 3-epoch acceleration attribution; extension beyond Taylor Valley where REMA supports it; calibrated-uncertainty product release (O3). Deliverable: synthesis/prediction paper + public data products.

Risks & mitigations. (1) *Barlow's label tiles are access-gated* → mitigation: the public LTER centerlines serve as a stand-in; plan to request the tiles and, failing that, re-digitize a comparable training set (the FI's prior work shows label generation is in hand). (2) *REMA's uneven post-2014 coverage* → prioritize Taylor Valley (full 3-epoch coverage) for acceleration; treat other valleys as 2-epoch. (3) *Discharge gauge gaps* → ERA5/AMPS energy reanalysis fills spatial/temporal holes (a core reason O1 uses melt energy, not just gauges). (4) *Outside the channels, change may sit below the detection threshold* → then O2 narrows back to in-channel change and the landscape-wide result is reported as a calibrated null — itself a quantitative statement about how spatially confined active change currently is.

Data Management Plan. All input data are free/public (NASA ATM, NCALM/OpenTopography, PGC REMA, MCM-LTER/EDI, Copernicus ERA5, AMPS/GDEX), cited per source. Derived products, change-detection rasters, per-stream rate tables, process-classified change maps, calibrated-uncertainty layers, and stream-boundary polygons, will be released open-access (DoIs via Zenodo / EDI) with the processing code (PDAL/WhiteboxTools/GDAL + Python). Heavy regenerable rasters are excluded from version control; lightweight scripts, figures, and tables are tracked. Outputs follow ASPRS/OGC standards and carry explicit CRS, method, and uncertainty metadata (per NASA open-science/trustworthy-EO guidance).

References

[1] Barlow, M. C. (2026). *A Comprehensive Spatial Analysis of Stream Boundary and Geomorphological Change Detection: McMurdo Dry Valleys, Antarctica.* PhD Dissertation, Dept. of Civil & Environmental Engineering, University of Houston (chair: C. L. Glennie). 240 pp.

[2] Barlow, M. C., Zhu, X., & Glennie, C. L. (2022). Stream Boundary Detection of a Hyper-Arid, Polar Region Using a U-Net Architecture: Taylor Valley, Antarctica. *Remote Sensing* 14(1):234. doi:10.3390/rs14010234. *(Dissertation Chapter 4, proof of concept.)*

[3] Höhle, J., & Höhle, M. (2009). Accuracy assessment of digital elevation models by means of robust statistical methods. *ISPRS J. Photogramm. Remote Sens.* 64(4):398–406. *(NMAD / robust DEM error.)*

[4] Howat, I. M., et al. (2019). The Reference Elevation Model of Antarctica (REMA). *The Cryosphere* 13:665–674. *(REMA.)*

[5] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI* 234–241.

[6] McMurdo Dry Valleys LTER. Stream discharge, meteorology, glacier mass-balance, and soil datasets, Environmental Data Initiative (EDI), `knb-lter-mcm.*`.

[7] Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146:1999–2049.

Draft S/T/M section, written as a plan only (no preliminary work presented). Not a submitted proposal.